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A LiDAR framework to characterise elevated and canopy firebrand fuels for fire prediction models

Ha-Ninh Nguyen[1,2], Trent D. Penman[1], Mika Peace[3], Nina Hinko-Najera[1], Sandra Penman[1], Jean-Baptiste Filippi[2], Alberto Alonso Pinar[1,2] and Alexander I. Filkov[1*]

Abstract

Background Firebrands, small burning particles, are a key driver of wildfire spread. Generated from burning vegetation, they are carried by wind to ignite spot fires ahead of the main fire front. Fire behavior models aim to predict this spread, but limited data on firebrand properties introduces uncertainty. Developing a fast and reliable method for firebrand fuel characterization is essential. Our study uses Terrestrial Laser Scanning (TLS) data to quantify the fine fuels in near-surface, elevated, and canopy forest layers. TLS scans were collected from three different temperate eucalypt forest sites in Victoria, Australia. A set of algorithms was used to extract and build characteristics of firebrand fuel from the TLS point cloud data. We categorized firebrand fuel into three types: twigs (diameter < 5 cm), candlebark and stringybark.

Results We were able to estimate variation in firebrand fuel in 1 m^3. The mean of twig length (diameter < 5.5 cm) was found to be 4.9–7.1 cm in the total plot, 1.5–1.7 cm in the canopy layer, and 15.4–24.8 cm in the elevated layer. The mean of candlebark length was 0.5–4.0 cm in the total plot, 0.2–1.0 cm in the canopy, and 2.9–13.6 cm in the elevated layer. The mean of stringybark surface area was 1218 cm^2 in the elevated layer and 344 cm^2 in the total layer. Fuel from different sites had varying characteristics. Our results indicated the overall trend across the three sites was for a decrease in twig length with increasing diameter, with the exception of candlebark, for which bark length decreased with increasing width. Fuel distribution was found to be highest in the near-surface and elevated layers and lowest in the canopy layer.

Conclusions The method we developed can be used as a new approach for quantifying fine fuel and hazard assessment. Collected data is especially valuable for laboratory-scale experiments, where controlled and replicable conditions are critical for studying firebrand generation and behavior. They can also serve as inputs to fire behavior simulation models, particularly for surface and crown fire models. Future research efforts could focus on enhancing the accuracy of algorithms and their implementation to further improve the effectiveness of fuel quantification.

Keywords Wildland fire, Firebrand fuel, Twigs, Candlebark, Ribbon bark, Stringybark, Fine fuel, Quantification , Terrestrial laser scanning, Quantitative structure models

*Correspondence:
Alexander I. Filkov
alexander.filkov@unimelb.edu.au
Full list of author information is available at the end of the article

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Resumen

Antecedentes Las pavesas --pequeñas partículas incandescentes producto de incendios de vegetación--, son los conductores claves en la propagación del fuego. Generadas por vegetación ardiente, estas pavesas son conducidas por el viento e inician fuegos por delante del frente de incendio. Los modelos de comportamiento del fuego se orientan a predecir esta propagación, aunque datos limitados sobre las propiedades de estas pavesas introducen incertidumbres. Por ese motivo, el desarrollar un método rápido y confiable para la caracterización de estas pavesas resulta esencial. Nuestro estudio usó datos del escaneo con laser Terrestre (TLS) para cuantificar los combustibles finos cercanos a la superficie, elevados, y correspondientes a capas del dosel arbóreo. Los escaneos con TLS fueron colectados en tres sitios diferentes de bosques templados de eucaliptus en Victoria, Australia. Un conjunto de algoritmos fue usado para extraer y construir las características de las pavesas basados en los datos de las nubes de puntos del escaneo. Categorizamos los combustibles que forman pavesas en tres tipos: pequeños tallitos (< 5 cm), pavesas producto del desprendimiento de corteza (*candlebark*) y corteza-filamento del eucaliptus (*stringybark*).

Resultados Fuimos capaces de estimar la variación de los combustibles (en 1 m^3), que una vez encendidos, forman pavesas. La media de los pequeños tallitos (diámetro menor a < 5.5 cm), fue de 4,9 a 7,1 cm en el total de la parcela, 1,5 a 1,7 en la capa intermedia del dosel arbóreo, y 15,4 a 24,8 cm en el dosel elevado. La media en la longitud del desprendimiento de corteza (*candlebark*) fue de 0,5 a 4,0 cm el total de la parcela, 0,2 a 1,0 cm en el dosel intermedio, 2,9 a 13,6 cm en el dosel elevado. La media en el área superficial de la corteza filamentosa fue de 1,218 cm^2 en el dosel elevado y de 344 cm^2 en el total de esta capa. El combustible de sitios diferentes presentó distintas características. Nuestros resultados indican que la tendencia general a través de los tres sitios resultó en un decrecimiento en la longitud de los tallitos pequeños al incrementarse su diámetro, con la excepción del *candlebark*, para el cual su longitud decreció al aumentar su diámetro. La distribución del combustible fue mayor en las cercanías de la superficie y en las capas elevadas del dosel y menor en capas intermedias del dosel.

Conclusiones El método que desarrollamos puede ser usado como una nueva aproximación para cuantificar los combustibles finos y determinar el peligro de incendio. Los datos colectados son especialmente valiosos para experimentos a escala de laboratorio, donde condiciones controladas y replicables son críticas para el estudiar la generación y el comportamiento de pavesas. Pueden también servir como componentes de los modelos de simulación, especialmente como modelos de fuegos de superficie y de copas. Los esfuerzos de investigaciones futuras podrían enfocarse en aumentar la exactitud de los algoritmos y su implementación, para mejorar la efectividad en la cuantificación de los combustibles.

Background

Fire regimes globally have shifted towards more intense and frequent fires in recent decades (Enright et al. 2015; Filkov et al. 2023a; Nolan et al. 2020). The change is driven by a combination of factors, including climate change, prolonged droughts, higher temperatures and altered precipitation patterns (Abatzoglou and Williams 2016; Cassell et al. 2019). Changes in land use, including deforestation and the expansion of urban areas, fragment ecosystems and increase in the Wildland–Urban interface (WUI) both fire risk and fire management complexity (Moreira et al. 2020). Additionally, fuel accumulation resulting from decades of fire suppression policy has created landscapes with excessive combustible materials, which intensify fire behavior (Miller et al. 2009). Invasive species, such as exotic grasses, also contribute to high fire loads by creating continuous fine fuels favorable for rapid fire spread (Adams 2013; Wotton et al. 2017). Together, these elements create conditions conducive to more severe and frequent fires. This poses a challenge for ecosystem management and human safety. To predict, prevent, and mitigate the negative impact of fire, we need to deepen our understanding of wildfire behavior, particularly the mechanisms that drive their spread.

Wildfire spread mechanisms include convection, radiation, and new ignitions (spot fires) by firebrands (Filkov et al. 2023b). Firebrands are burning particles consisting of twigs, bark, cones, and other small organic debris (McArthur 1961; Peet 1965). Spot fires are associated with the majority of structure losses and lives lost during wildfires (Manzello et al. 2006; Wadhwani et al. 2022). The spotting process consists of three key stages: generation, transport, and fuel ignition (Koo et al. 2010). Firebrands, once generated in the fire front, can be lofted in the fire plume and transported far ahead of the fire front by wind (short-range spotting) or lofted by turbulence near the convective column (medium- and long-range spotting) (Koo et al. 2010; Manzello et al. 2006; Sardoy et al. 2007). Once landed, they can start new fires. Improved knowledge of firebrand processes including generation, transport, fuel ignition, and the inclusion of

these processes in fire spread models is critical for accurate predictions of fire propagation.

Firebrand fuel, which is part of fine fuel, generally consists of twigs and bark (Manzello et al. 2007, 2022) distributed across different fuel strata within a forest (e.g., Fig. 1). The lower layer, which includes surface, near-surface and elevated fuel, is typically burned in surface fires, while the canopy layer burns in crown fires (Hines et al. 2010). Fuel characteristics and loads vary widely across woody plant species (Pausas 2015), and these layers contain diverse types and amounts of fine fuel, which influence fire propagation (Roos and Scott 2018). Understanding how the different layers contribute to generation and quantity of firebrands is crucial to produce accurate predictions of spotting during wildfires. Current spotting models (Kepert et al. 2024; Koo et al. 2012; Sardoy et al. 2007; Tolhurst et al. 2008) rely on limited case studies or experiments (e.g., Gould et al. 2008b; Hudson and Blunck 2019; Koo et al. 2010; McCaw et al. 2012; Suzuki et al. 2012; Wang et al. 2011). Moreover, there is a lack of data on the quantity of generated firebrands (El Houssami et al. 2016; Filkov and Prohanov 2019). Accurate input data on firebrand characteristics (including number, size, shape, area and volume) are essential for improving model accuracy (Alonso-Pinar et al. 2025).

The characteristics of firebrands can be described using various methods. Traditionally, fuel characteristics have been described using destructive measurement methods, including field sampling, oven drying, and weighing biomass per unit area. These methods particularly focus on sampling of twigs, bark, leaves, and cones, as well as describing parameters such as fuel bed depth, percentage coverage, fuel continuity and arrangement in horizontal layers (Gould et al. 2011; McArthur 1961; Peet 1965; Volkova and Weston 2013). While direct measurements are relatively accurate, they are labor-intensive, time-consuming, and present significant challenges in inaccessible terrains and dense forests (Alvarez et al. 2012; Mani and Parthasarathy 2007; Ravindranath and Ostwald 2008).

Alternative approaches using remote sensing techniques are becoming increasingly popular for vegetation measurements. Light Detection and Ranging (LiDAR) uses laser beam technology to measure the distance and shape of objects in the environment (Gleason and Im 2012). Over the past decade, LiDAR has been used to reconstruct 3D spatial forest structures, allowing researchers to develop algorithms to analyze the LiDAR data and accurately extract vegetation properties (Lefsky et al. 2002; Lim et al. 2003).

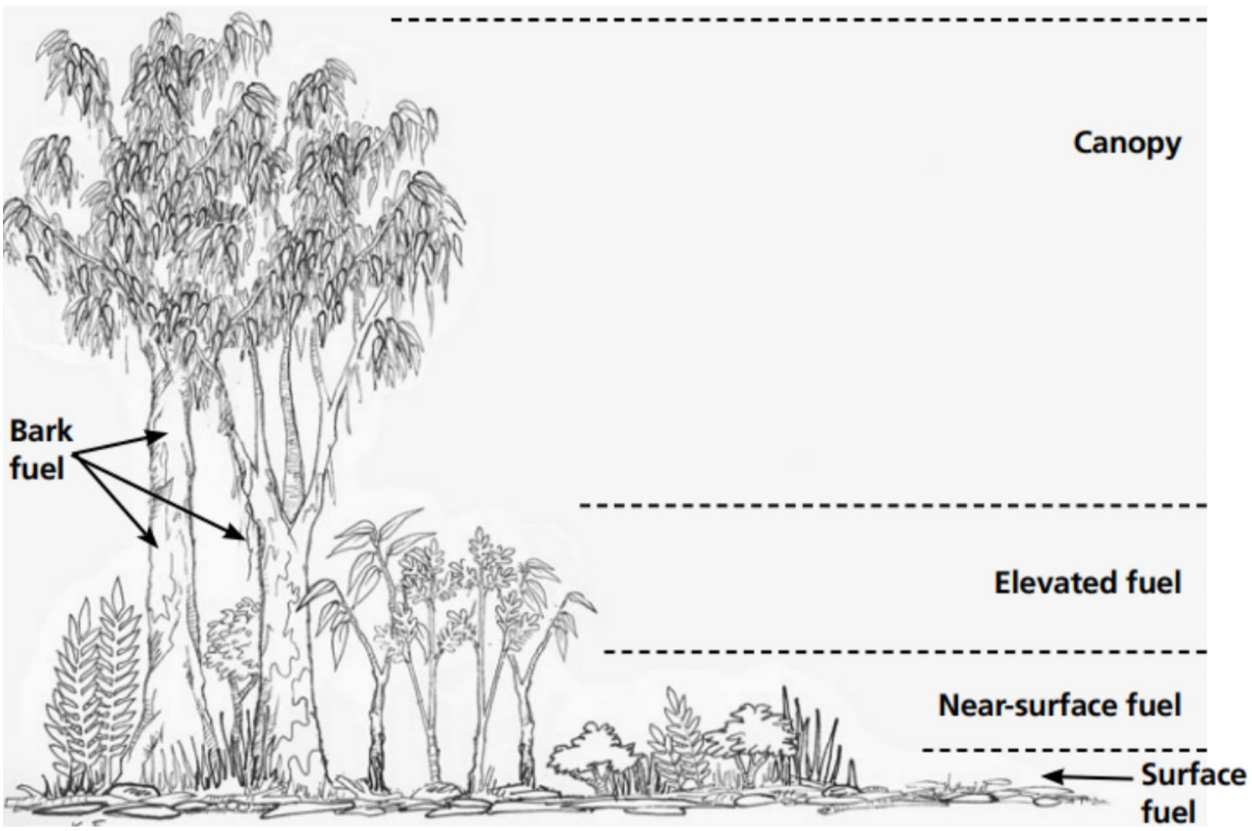


Fig. 1 Diagram showing typical fuel layers (adapted from Hines et al. 2010)

One commonly used method for collecting LiDAR data is ground-based Terrestrial Laser Scanning (TLS). It is becoming an increasingly popular method for estimating Above Ground Biomass (AGB) (Calders et al. 2015; Demol et al. 2022a; Gonzalez de Tanago et al. 2018). Validating AGB assessments obtained through TLS is crucial to ensure the reliability and accuracy of these measurements in real-world applications. These include evaluating the potential of forests to provide sustainable wood resources, assessing their role in carbon sequestration for climate change mitigation, and monitoring forest ecosystems for fuel and energy production (Malmsheimer et al. 2011).

A thorough evaluation of TLS performance across ten destructive-harvest studies was conducted by Demol et al. (2022a), in which TLS-derived AGB estimates were systematically compared against conventional Allometric Scaling Models (ASMs). These ASMs primarily represent whole-tree AGB formulations parameterized using stem-centric predictors—most commonly "Diameter at Breast Heigh" (DBH), and, where available, tree height and wood basic density—and implicitly account for twig, branch, and foliage biomass through empirically fitted scaling relationships (Picard et al. 2012). For example, pantropical ASMs based on the framework of Chave et al. (2014), integrating DBH2·height·wood density terms, were used in Burt et al. (2021); Gonzalez de Tanago et al. (2018); Lau et al. (2019); Momo Takoudjou et al. (2018). Species-specific DBH-based power-law models, calibrated to represent total above-ground woody biomass (trunk + branches, with foliage implicitly included), were used by Calders et al. (2015); Demol et al. (2021); Hackenberg et al. (2015) while Stovall et al. (2017) used genus-specific ASMs of similar structure. In general, their findings indicate that TLS-derived AGB shows a bias of less than 1% (where lower bias is better) and a concordance correlation coefficient of 98% (where higher correlation is better), underscoring the reliability of this method. Notably, for large trees, TLS provides more accurate and less biased results compared to ASMs (Demol et al. 2022a).

Commonly, exported TLS data for analysis is called a point-cloud, a discrete set of data points in space. The points represent tree shapes and forest structures. Several algorithms have been developed for reconstructing 3D tree models from point cloud data, including AdQSM (Dong et al. 2021; Fan et al. 2020), AdTree (Du et al. 2019), PypeTree (Delagrange et al. 2014), SimpleTree (Hackenberg et al. 2015a) and Outer Hull Modelling with voxelization (Stovall et al. 2017). These algorithms have generally demonstrated close agreement with data from destructive forest sampling, although they often employ different verification indices. As a result, there has not yet been a comprehensive study that objectively compares the different algorithms. Nevertheless, 9 out of 10 studies employed Quantitative Structure Models (QSMs) for estimating AGB, with 7 of the 9 studies using TreeQSM (Demol et al. 2022a). This widespread use of TreeQSM can be attributed to its frequent code updates and excellent performance since its introduction.

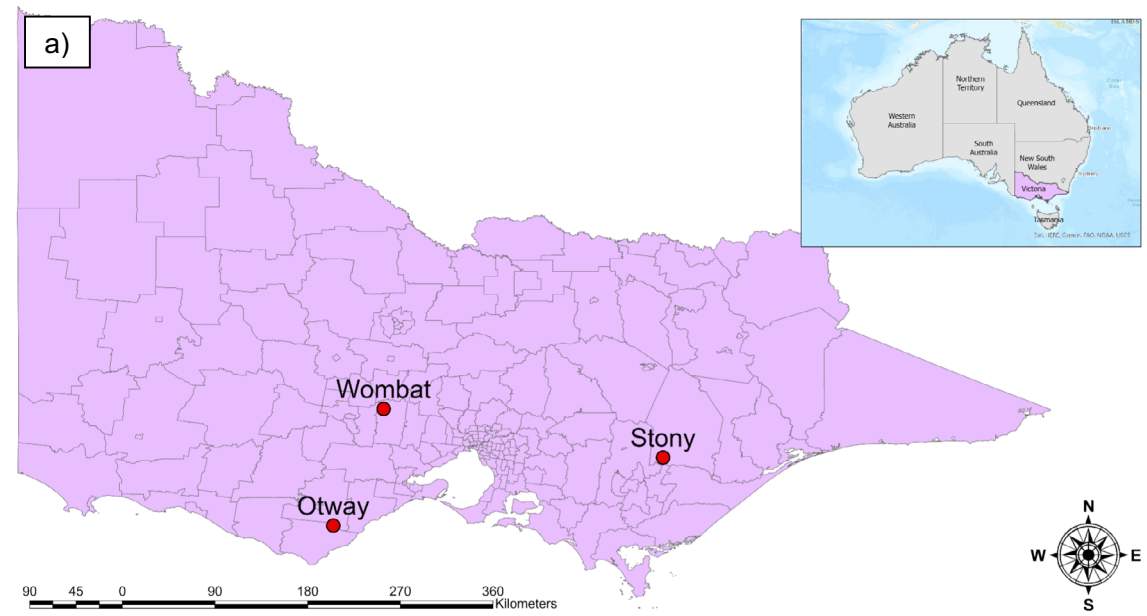
Previous studies have highlighted the advantages of using LiDAR technology for quantifying forest fuel through canopy height and "Canopy Base Height" (CBH, commonly defined as the vertical distance between the ground surface and the base of the lowest continuous live canopy) assessments (Andersen et al. 2005; Chen et al. 2016; Erdody and Moskal 2010; Gleason and Im 2012; Jakubowksi et al. 2013; Lim et al. 2003). However, these studies have not categorized fuels based on their type (e.g., twigs and bark), size, fuel layer, and forest types, thereby limiting the detail of fuel characterization. This research aims to address this gap by proposing advanced TLS methods to quantify fine fuels, specifically twigs and bark across various fuel layers and forest types, with a particular focus on firebrand generation applications.

Proposed method

Study area

Because conducting new TLS scans was outside the scope of this study, we utilized existing TLS datasets collected across different temperate eucalypt forest types in Victoria, Australia. The three selected sites—Great Otway National Park, Wombat State Forest, and Stony State Forest (Fig. 2)—were chosen to represent a range of forest types, dominant species, and geographical locations. This diversity allowed us to test the proposed framework across different vegetation structures and fuel characteristics. The datasets cover approximately 1.3 hectares of vegetation divided into 29 plots, with plot sizes ranging from 60.68 to 1018 m^2 (Appendix A). These sites span temperate regions with warm, dry summers and cool, wet winters (Harris et al. 2015; Mills et al. 2020).

Otway is situated in the Victorian Otway Ranges bioregion. All plots in this site are within the Foothills Forest vegetation class (Penman et al. 2023). This area is characterized by both dry and wet sclerophyll forests with a mixed canopy predominantly composed of *Eucalyptus* species: *E. obliqua, E. cypellocarpa*, *E. globulus* (bluegum), and *E. viminalis* (manna gum) (Penman et al. 2023). The understory layer is characterized by dense shrubs, primarily composed of wire grass (*Tetrarrhena juncea*), which becomes highly flammable during the summer months and is associated with moderate-to-low frequency, high-severity fires (Leonard et al. 2016; McColl-Gausden and Penman 2019; Swan et al. 2015).



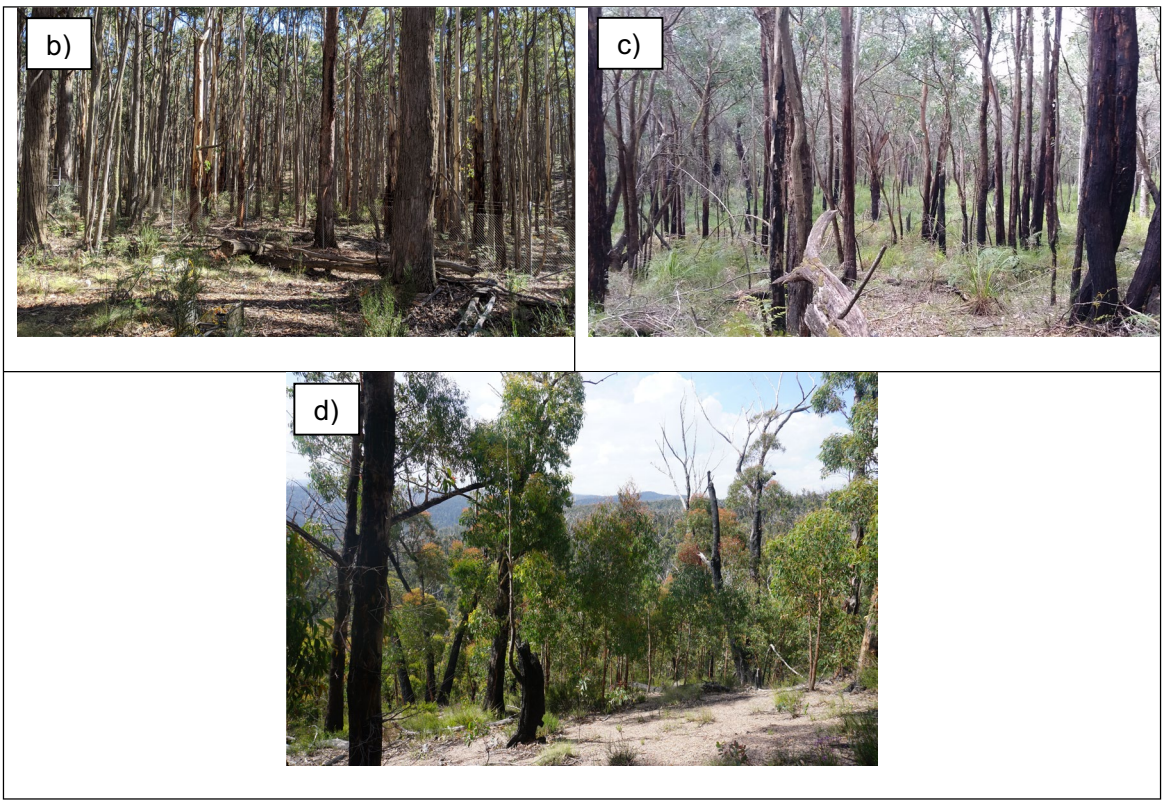


Fig. 2 Location and representative vegetation of the three TLS study sites in Victoria, Australia. **a** Map showing the locations of Wombat State Forest (Wombat hereafter), Great Otway National Park (Otway hereafter), and Stony State Forest (Stony hereafter). **b** Representative vegetation at Wombat. **c** Representative vegetation at Otway. **d** Representative vegetation at Stony

Wombat is located in central Victoria in the Goldfields region and is characterized as a dry sclerophyll open forest dominated by fire-tolerant mixed eucalypt species: *E. obliqua*, *E. rubida*, and *E. radiata*. The understory is sparse with predominantly Austral bracken and perennial grasses. The forest has been occasionally burned by wildfire and has been managed with regular low-intensity prescribed fires. The selected sites are secondary regrowth forests and were last selectively harvested in the early 1970 s, with the last wildfire on the outskirts of the study site recorded in 1982, no recorded history of prescribed fires (Hinko-Najera et al. 2017, 2019; Plumanns-Pouton et al. 2022). Although the Wombat plots have been subject to windthrow, TLS scans used in this study were collected after the 2021 windthrow event, and only from locations that experienced low-severity impacts (Hinko-Najera et al. 2024). Plot size was generally of 15 m radius with 4 scan positions at roughly cardinal points.

Stony site is located in Stony and Deep Creek State Forests, which fall within the Victorian Highlands–southern fall bioregion. The sites were all located in dry forest which was interspersed with wetter gullies containing wet and damp sclerophyll forests. The dominant eucalypt species across the study sites are *E. consideniana*, *E. sieberi*, *E. muelleriana*, and *E. tricarpa* (Collins 2020). The area is known for its significant variation in topography and microclimates, which contribute to diverse vegetation structures.

Scans were conducted in dry weather and no wind conditions to ensure accuracy in measuring vegetation. More details about the plots can be found in Appendix A.

Data collection and processing

Firebrand fuel can be categorized into three main types: twig fuel, stringybark fuel, and candlebark (also known as "ribbon bark") fuel (Fig. 3). Each of these categories plays a crucial role in the generation and transport of firebrands during wildfires. We used different approaches to define and analyze each type of firebrand fuel.

Existing literature suggests that twigs exceeding 5 cm diameter are not usually consumed in wildfires and are less likely to contribute to firebrand formation (El Houssami et al. 2016; Filkov et al. 2017, 2023b; Gould et al. 2008a; Manzello et al. 2007). Due to the limited ability of TLS to reliably detect twigs less than 0.5 cm in diameter, these fuels were not considered in this analysis. Accordingly, only twigs with diameters > 0.5 and < 5 cm were included in the assessment of firebrand fuel. While fine fuels (< 0.5 cm diameter) are widely recognized as important contributors to fire dynamics and are critical for modeling fire front propagation, the focus of this study is on fuels relevant to firebrand generation rather than those primarily driving the advance of the main fire front. Fine fuels below 0.5 cm in diameter are typically fully consumed within the flaming zone and are therefore unlikely to constitute a significant source of firebrands. As a result, their exclusion is not expected to substantially influence estimates of above-ground firebrand fuel. Similarly, although surface litter plays an important role in sustaining and propagating the flaming front, it is generally not considered a primary contributor to firebrand generation and was therefore excluded from the firebrand fuel assessment.

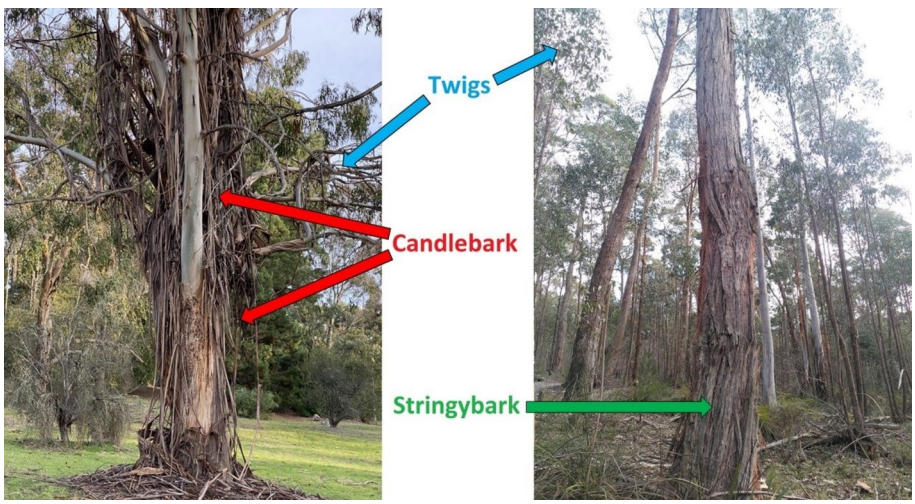


Fig. 3 Three main firebrand fuel types (twigs, stringybark, and candlebark). The left image shows a candlebark tree, and the right image shows a stringybark tree. Twigs are represented in both types of trees

Stringybark fuel is characterized by its fibrous and peeling bark structure, which is recognized as the main source of firebrands in Australia (Chattaway 1955; Ellis 2011, 2013). The stringybark can easily detach from the tree during a fire, creating firebrands that can be carried by wind over long distances (Ellis 2011; Penman et al. 2017). Stringybark fuel is particularly common in certain species of *Eucalyptus*, for example, *E. obliqua*, which are known for their flammable bark. To define stringybark, we extracted the total trunk area of all trees within a plot from TLS data.

Candlebark fuel refers to the long stripes of bark resembling candles that hang from gum-type eucalypt trees (Fig. 3). During a wildfire, candlebark can be ignited and can produce large firebrands that contribute to spotting and fire spread. In this study, we developed a method to detect candlebark on a tree and quantify its length and width (see Section "Data analysis" for more details).

The workflow for our firebrand fuel characterization method is illustrated in Fig. 4. The flowchart demonstrates the process of analyzing TLS data to detect and quantify forest fuel, applied to individual trees within a scanned area. The process begins with TLS data collection, involving detailed 3D scans from multiple positions to ensure comprehensive coverage. Following this, the individual raw scans undergo a registration step to merge them into a single point cloud dataset. The point cloud is then segmented to isolate individual trees using the Treeiso algorithm (Landrieu and Obozinski 2017). Each segmented tree is reconstructed into a detailed 3D model with TreeQSM, enabling analysis of physical characteristics such as trunk and twig dimensions. The output data, including trunk and branch metrics, is used to derive various forest structure metrics, such as elevated fuel, canopy fuel, and bark type. The final step converts these metrics into fuel amount per cubic meter for comparison, estimation, and further analysis. A detailed description of each step is presented below.

Data collection

The 3D point cloud data for the Otway sites were sourced from previously published research (Penman et al. 2023), comprising 24 plots scanned using a FARO Focus 3DS LiDAR in outdoor collection mode (resolution 1.6 mm at 10 m). The remaining 5 plots (Wombat and Stony) were scanned with RIEGL VZ-400i LiDAR with a comparable setting (scan pattern was Pano 40, 100° angle, and laser pulse repetition rate was 600 kHz). A single scan size was approximately 300 M points for each scanner. Multiple scan positions were conducted for each plot to capture the study area. The distances between the scan positions varied to minimize obstruction from large trees and dense shrubs (Fig. 5).

Data processing

Following the processing steps outlined for the Otway dataset by Penman et al. (2023), we replicated the same downstream processing and analysis pipeline for the Wombat and Stony sites. This included registering individual scans together and applying ground normalization and cropping using the authors' R script. All processing was conducted using RiSCAN Pro software (version 2.16.1; RIEGL Laser Measurement Systems GmbH, www.riegl.com). Specifically, after the area of interest within the scan positions was extracted to a LAS/LAZ file (a format for three-dimensional data files), we ran the ground normalization. No explicit filtering or removal of foliage or fine non-woody vegetation was applied prior to TreeQSM reconstruction. The TLS point clouds therefore represent a combination of woody structure and retained foliage. While TreeQSM (Section "Tree reconstruction") is designed to reconstruct woody architecture, we recognize that the presence of foliage and fine vegetation may locally affect point density and occlusion, resulting in biomass overestimation (Luck et al. 2023). We used CloudCompare with a SOR filter (NN = 10, Std = 1) and CFS filter to clean the point cloud and for ground and noise removal (Zhang et al. 2016).

Individual tree segmentation

Tree segmentation is the process of delineating trees from the point cloud. The TreeQSM algorithm is capable of reconstructing one tree at a time (Raumonen et al. 2013, 2015), which makes it necessary to apply it separately to each tree in all plots. To avoid manual segmentation, another algorithm, Treeiso, was used to isolate individual trees. It was introduced by Landrieu and Obozinski (2017) to segment a single tree from the whole plot TLS scan automatically, also used by Raguet and Landrieu (2018) and Xi and Hopkinson (2022).

Tree reconstruction

We used TreeQSM to reconstruct the point cloud data into a 3D model (Markku et al. 2015; Raumonen et al. 2013, 2015) in MATLAB and analyze the trunk area, length and diameter of twigs, as well as length and width of candlebark. The output of TreeQSM is individual tree point cloud data with two files: branch structure and comprehensive tree metrics. The "branch structure" file contains various parameters such as branch order, parental connection, dimensions (diameter, volume, area, length, height), and orientation (angle, azimuth, zenith). The "Treedata" file includes total, trunk, and branch volumes; tree trunk, and branch dimensions; count and structural data of branches; dendrometric parameters like DBH; and crown metrics (such as average and maximum diameter, base height, volumes, etc.).

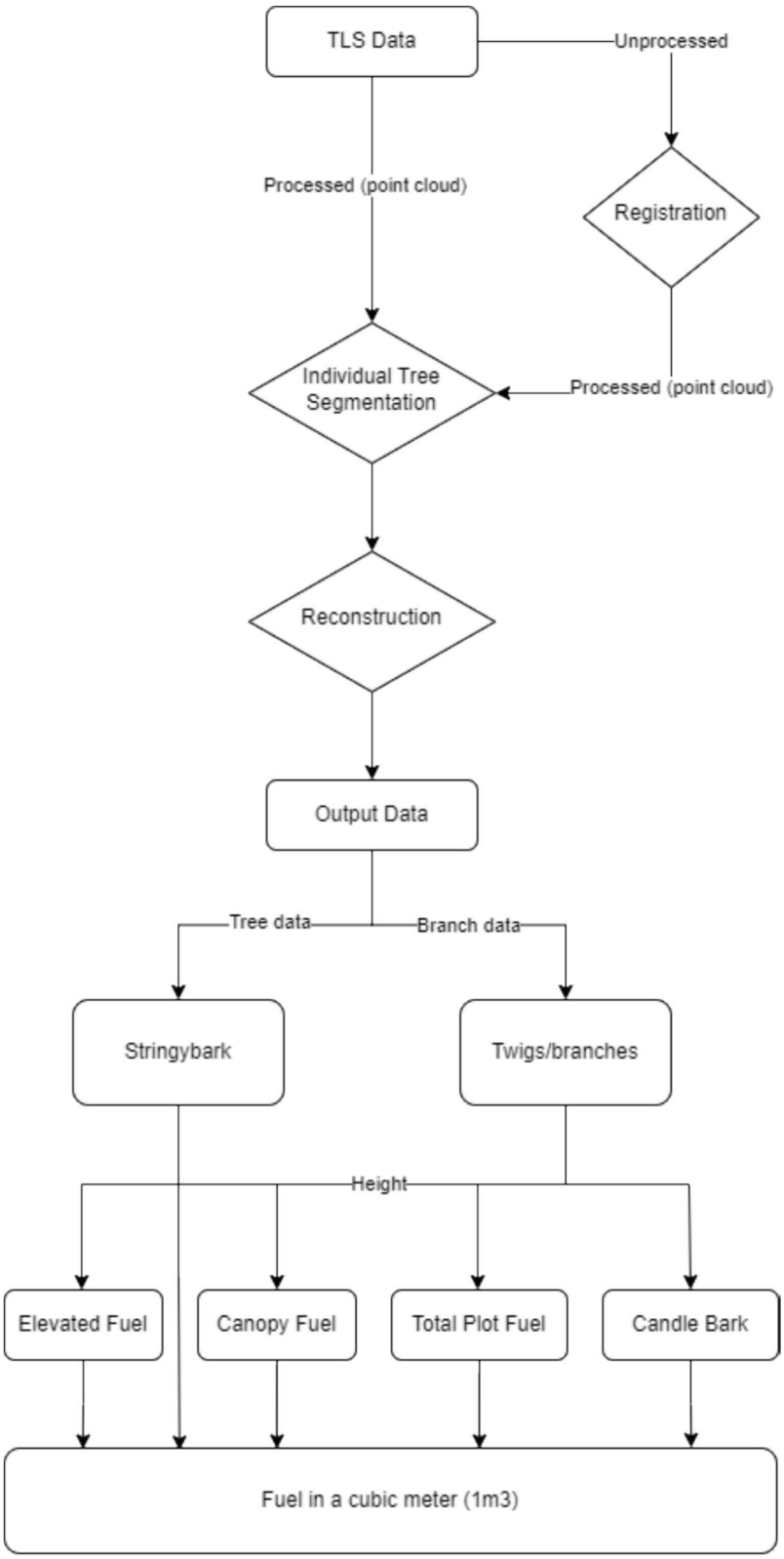


Fig. 4 Workflow for TLS-based firebrand fuel characterization and quantification. The process begins with TLS data collection and registration of raw scans into a processed point cloud, followed by individual tree segmentation using Treeiso and 3D tree reconstruction using TreeQSM. Output tree and branch data are then used to classify firebrand fuel types, including stringybark, twigs/branches, and candlebark, across elevated, canopy, and total plot fuel layers. The final output expresses firebrand fuel quantity per cubic meter (1 m^3)

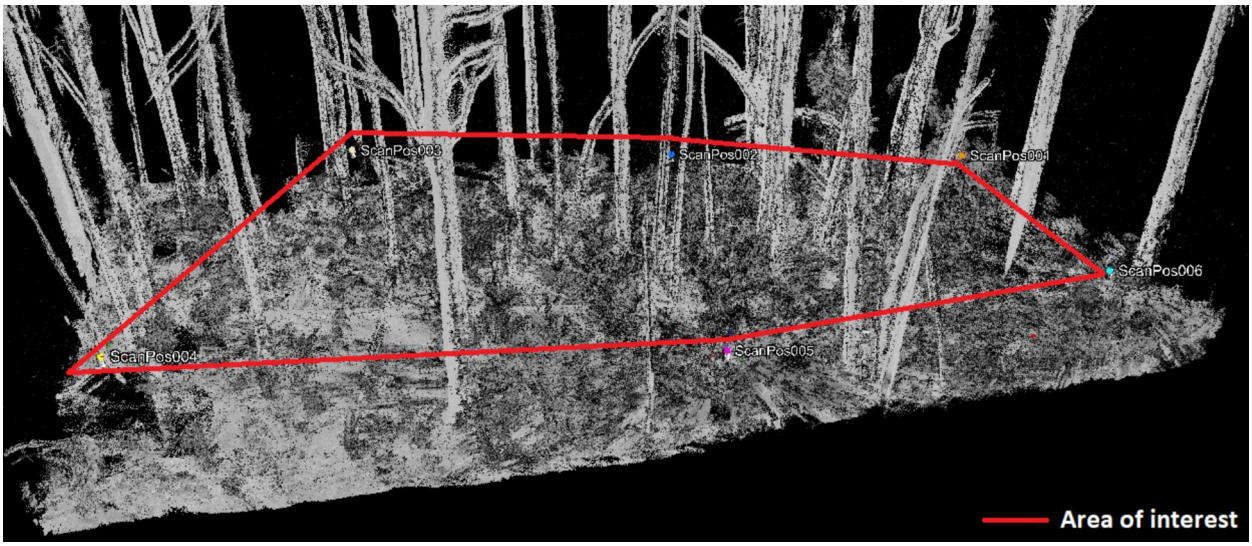


Fig. 5 The area of interest and scan positions for one example plot. The area of interest is typically in the middle of the scan positions to be able to maximise the detail of all tree angles

Derived metrics

In our study, we specifically analyzed the metrics: "diameter" (m), "length" (m), and "height" (m) for twigs; and "trunk length" (m) and "trunk area" (m^2) for stringybark species (Appendix B). These metrics are generated by TreeQSM.

Surface fires generally burn fine fuel in the surface, near-surface, and elevated fuel layers, while crown fires burn fuels across all layers, including the canopy layer (Fig. 1). This distinction is critical because different fire types interact with different vertical strata of fuel. Surface fuel (litter) was not considered in our analysis as it does not generally contribute to firebrand generation (Filkov et al. 2016) and cannot be reliably estimated using TLS. We combined the near-surface and elevated layers into a single layer referred to as the elevated layer.

Our selective approach allowed us to categorize the twig distribution across different vertical strata within the plot—elevated, canopy, and total fuel layers. By categorizing the twigs into these layers, we can more accurately model and predict the firebrand behavior.

Data analysis

Twigs The classification scheme of firebrand fuel used in our analysis is detailed below, outlining the dimension categorization and associated twig location within the forest structure. Twigs are categorized into five diameter classes: 0.51–1.5 cm, 1.51–2.5 cm, 2.51–3.5 cm, 3.51–4.5 cm, and 4.51–5.5 cm. Additionally, height is used as a criterion to categorize twigs based on their vertical position within the plot. This categorization distinguishes between the overall forest height, the canopy layer, and the elevated fuel layer, which is defined as being below 2.5 m (Cheney et al. 2012; Cruz et al. 2021) (Fig. 6).

Recent TLS-based fuel quantification studies have primarily focused on general structural descriptors or aggregate fuel metrics, with few explicitly quantifying firebrand-relevant fine fuels by fuel type and vertical fuel layer within a standardized volumetric framework (Calders et al. 2020; Demol et al. 2022b; Forbes et al. 2022; García et al. 2011; Hillman et al. 2021; Hudak et al. 2020; Loudermilk et al. 2023; Pokswinski et al. 2020; Ross et al. 2022; Rowell et al. 2020; Terryn et al. 2024). To address this gap, we propose reframing wildfire fuel estimation as an engineering problem by standardizing measurements within a defined 1 m^3 volume. This approach provides a consistent, replicable framework for studying firebrand generation and facilitates scaling to larger forest areas, enabling more precise estimates of potential firebrand production during wildfires.

When firebrand fuel spanned multiple vertical layers, it was allocated between layers, with all material below 2.5 m assigned to the elevated layer and all material above the CBH assigned to the canopy layer. To estimate fuel load per cubic meter (1 m^3) for each layer, the total amount of fuel within that layer was calculated and then divided by the corresponding layer volume. The dimensions of each plot (length, width, canopy height, and CBH) are defined using the measure tool in

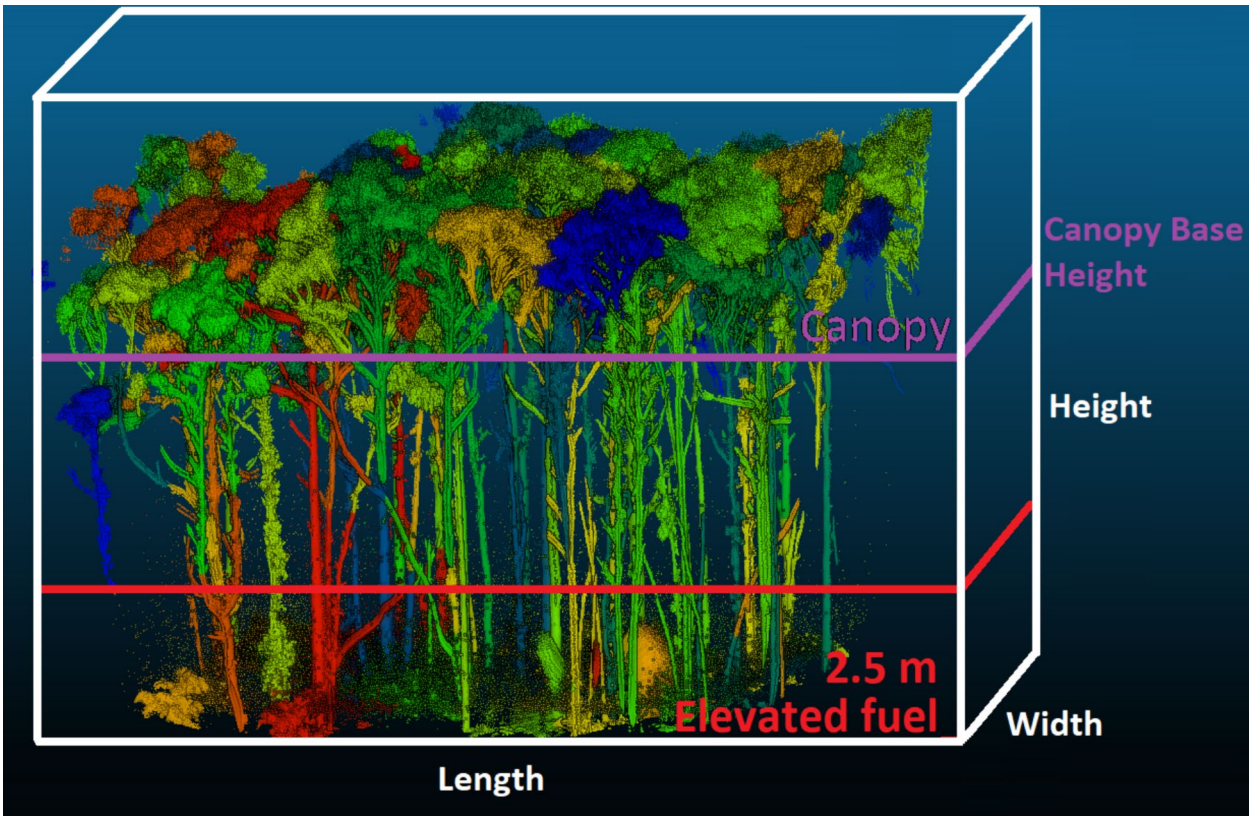


Fig. 6 Fuel TLS-based characterization. Individual trees were segmented and visualized using distinct colors. Fuel layers were defined based on height and classified into Canopy, Elevated, and Total layers. The volume of each layer was calculated by the length, width, and height of the plot

CloudCompare (values provided in Appendix A). The total volume of each layer was calculated as follows (Fig. 6):

$$V_{\text{elevated}} = a \times b \times 2.5 \tag{1}$$

$$V_{\text{canopy}} = a \times b \times (h_{\text{total}} - h_{\text{CBH}}) \tag{2}$$

$$V_{\text{total}} = a \times b \times h_{\text{total}} \tag{3}$$

where V_{elevated} is the volume of Elevated fuel layer (m^3), V_{canopy} is the volume of Canopy layer (m^3), V_{total} is the volume of the Total layer (m^3); a and b are the width and length of the plot (m), h_{total} is the height of the plot (m), h_{CBH} is the height of the canopy base (m) (Fig. 6).

The twig length in a cubic meter for each diameter was calculated using the following equation:

$$L_i = \frac{\text{length}_i}{V_{\text{layer}}} \tag{4}$$

where i is a twig diameter class, L_i is the length of twigs in 1 m^3 for each diameter (m), length_i is sum of twig length of each diameter class in a forest layer (m), and V_{layer} is the layer volume calculated using Eqs. (1) – (3) (m^3).

Stringybark In this study, stringybark area is calculated as the trunk area. The variation in trunk diameter along its height, typically larger near the base and tapering towards the top, creates significant challenges for accurate trunk area estimation using traditional methods. TreeQSM uses the point cloud data to reconstruct trees and allows precise characterization of trunk dimensions at various heights. This ensures a highly accurate estimation of the trunk area and addresses the limitations of other methods (Raumonen et al. 2013).

Stringybark area in a cubic meter is calculated using the following equations:

$$A_{cubic_el} = \frac{\sum\left(\frac{2.5}{L_{trunk}} \times A_{trunk}\right)}{V_{elevated}}, if\ L_{trunk} < 2.5\ m\ then\ A_{cubic_el} = \frac{\sum(A_{trunk})}{V_{elevated}} \tag{5}$$

$$A_{cubic_total} = \frac{\sum (A_{trunk})}{V_{total}} \quad (6)$$

where $A_{\text{cubic_el}}$ is the area of stringybark in 1 m^3 in the Elevated layer, L_{trunk} is the length of trunk (m), A_{trunk} is the area of the trunk (cm^2), and V_{elevated} and V_{total} are layer volumes calculated using Eqs. (1) and (3) (m^3).

It is important to note that bark depth, charring level, and degree of attachment are important characteristics of stringybark which would impact on the number of generated firebrands. To consider this, we recommend future studies using the "Overall fuel hazard assessment guide" from Australia (Hines et al. 2010). It separates stringybark trees based on the hazard rating: "extreme," "very high," "high," "moderate," which represent the amount of bark, charring level, and its degree of attachment.

Candlebark TreeQSM and other similar algorithms have not been previously used to characterize candlebark. However, our review of TreeQSM indicates that the algorithm is capable of detecting candlebark formations on trees. In our study, we used TreeQSM's capabilities to develop a method to characterize candlebark.

Candlebark usually hangs vertically downward and is hence considered to be at around a 90-degree angle relative to the ground. Within TreeQSM, the parameter "azimuth" is used to assess the orientation of each segment of the tree, including branches and potentially, candlebark formations. The azimuth angle ranges from −180 to 180°, where 0° aligns with the positive x-direction, 90° with the positive y-direction, −90° with the negative y-direction, and both 180 and −180° align with the negative x-direction.

Given that not all candlebark hangs perfectly straight down, there can be variations in its orientation. Therefore, our methodology includes a calibration approach to account for these irregularities. By establishing a baseline where a perfectly vertical (hanging) candlebark is defined at 0° azimuth, we tested deviations from this ideal condition at incremental angles: 15°, 20°, 25°, and 30° (Fig. 7). These angles were chosen to simulate the possible variations in the hanging angle of the candlebark due to natural factors or growth patterns.

To test the method, a calibration was performed by scanning a candlebark tree using TLS and manually collecting bark from it to compare the methodologies. The section for bark quantification (shown in Fig. 8) was limited by the maximum height accessible for manual data collection. This section was used for the comparison of bark characteristics between the manual measurements and the digital model represented in the point cloud data.

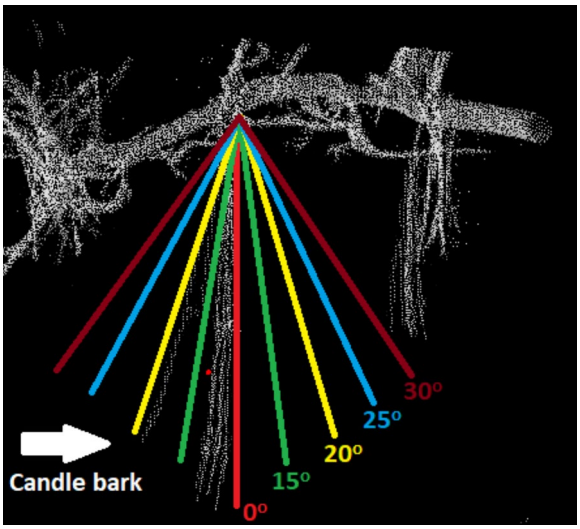


Fig. 7 Point-cloud image of a section of a candlebark tree. The small white points represent twigs and candlebark, while the coloured straight lines indicate angles measured relative to the vertical axis

A total of 21 bark samples were manually measured on the tree, with lengths varying from 1.16 to 4.34 m and widths ranging from 0.92 to 10.17 cm. The total length of all candlebark samples combined was 44.25 m.

Analysis of Relative Accuracy (RA, smaller is better) between the experimental measurement (L_{exp}) and calculated by TreeQSM (L_{QSM}) lengths at various angles indicates that the 20-degree angle provides the closest match to field measurements, 9% (Table 1). This angle was later used in our candlebark characterization.

In this study, we assumed that each plot contains only one bark type, based on the dominant species, either stringybark or candlebark. This assumption is justifiable given that our objective was to develop a framework for defining firebrand fuel rather than estimating precise fuel quantities for each plot. We acknowledge that, in most cases, plots may include both stringybark and candlebark trees. In such scenarios, bark firebrand fuel could be estimated proportionally based on the relative abundance of each tree type.

Results

There were different numbers of plots among the three study areas, with 24 plots in Otway, 4 in Wombat, and 1 in Stony. Given this imbalance, the results are presented primarily to demonstrate the performance of the framework rather than to support direct quantitative comparisons among sites. Firebrand fuel results are summarized by fuel component: twigs, candlebark, and stringybark.

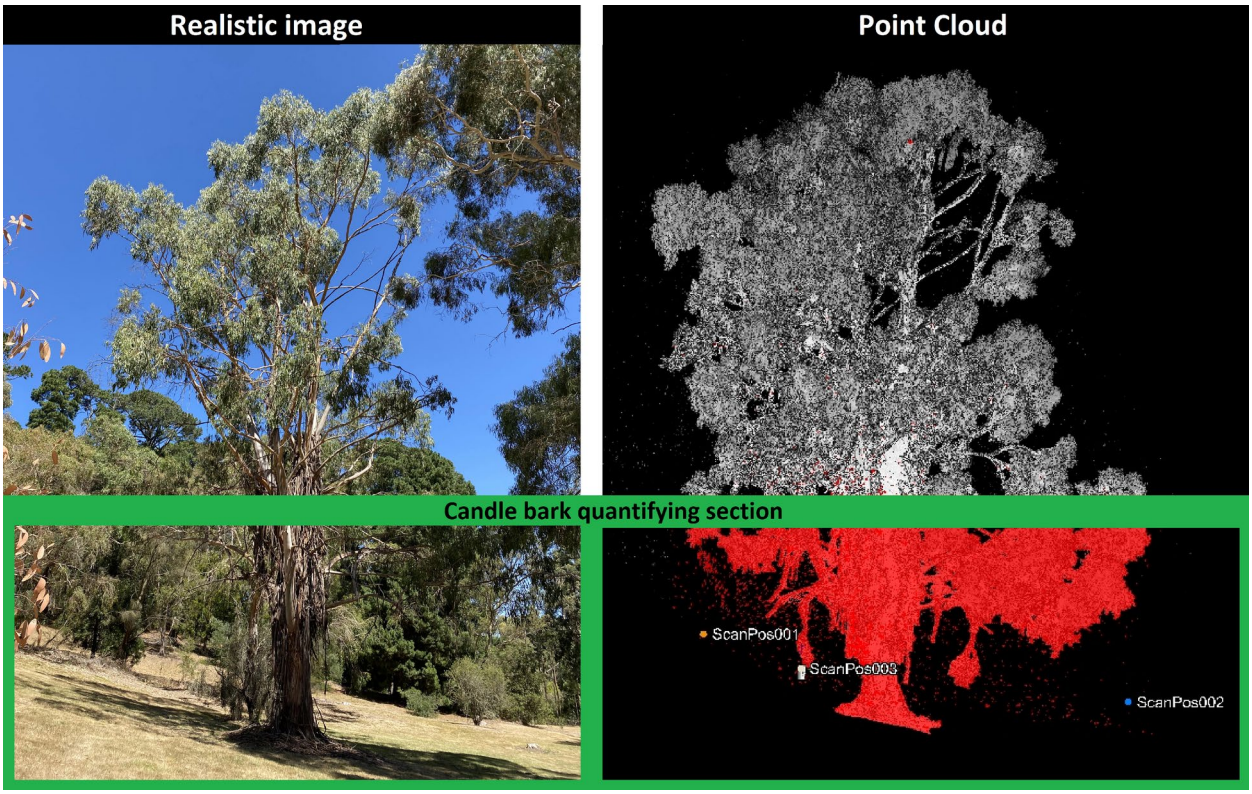


Fig. 8 Candlebark quantification. The section for destructive sampling and calibration is highlighted by a green rectangle

Table 1 RA between the field and algorithm measurements, where *Angle (deg)* is the angle of the candlebark shown in Fig. 7; L_{exp} *(m)* is total measured candlebark length; L_{QSM} *(m)* is the total candlebark length estimated by TreeQSM

Angle (deg)	L_{exp} (m)	L_{QSM} (m)	RA (%)
15	44.25	29.122	52%
20	44.25	40.622	9%
25	44.25	50.197	12%
30	44.25	57.917	24%

Twigs

Variations in twig length across different diameter classes and layers (Total, Canopy, Elevated) in 1 m^3 are shown in Figs. 9 and 10. The twig diameter is categorized into five classes ranging from 0.51 to 5.5 cm. The twig metrics include mean, median, quartiles, and range of diameter classes and length values.

Figure 9 compares twig lengths between the three sites: Otway, Wombat, and Stony. For Otway (S1), twig length does not vary significantly with twig diameter classes. The mean lengths are all less than 5 cm in the Total layer, less than 3 cm in the Canopy layer, and less than 30 cm in the Elevated layer. The maximum twig length across all twig diameters for the Otway is the lowest of the three sites, except for the 1 cm twig diameter of the Elevated layer, which is around 34 cm, while the maximum value of the Wombat site is 23 cm.

In contrast, the Wombat site (S2), which includes 4 plots, shows the most varied lengths. Mean lengths in the Total layer range from 15 to 24 cm, in the Canopy layer from 3.4 to 7 cm, and in the Elevated layer from 19 to 43 cm. There is considerable variation in twig lengths across twig diameter classes, with the mean length increasing from class 1 to class 3 diameter, then decreasing in class 4 to class 5 diameter.

A similar trend is observed at the Stony site (S3), although twig lengths are generally higher than those at the Wombat site. Across all diameter classes, Wombat (S2) typically has the highest maximum values compared with other sites. However, the 2 cm and 5 cm twig diameter in the Canopy layer are highest at the Stony site (S3). Twig diameters for class 2 and 5 in the Canopy layer at the Wombat site are approximately 7.6 cm and 7.7 cm respectively, lower than the maximum values of Stony (S3), which are 7.7 cm and 10.9 cm.

Figure 10 shows the twig diameter class data for all three sites. In the Elevated layer, mean twig length shows a slight increase from class 1 to class 2, then a

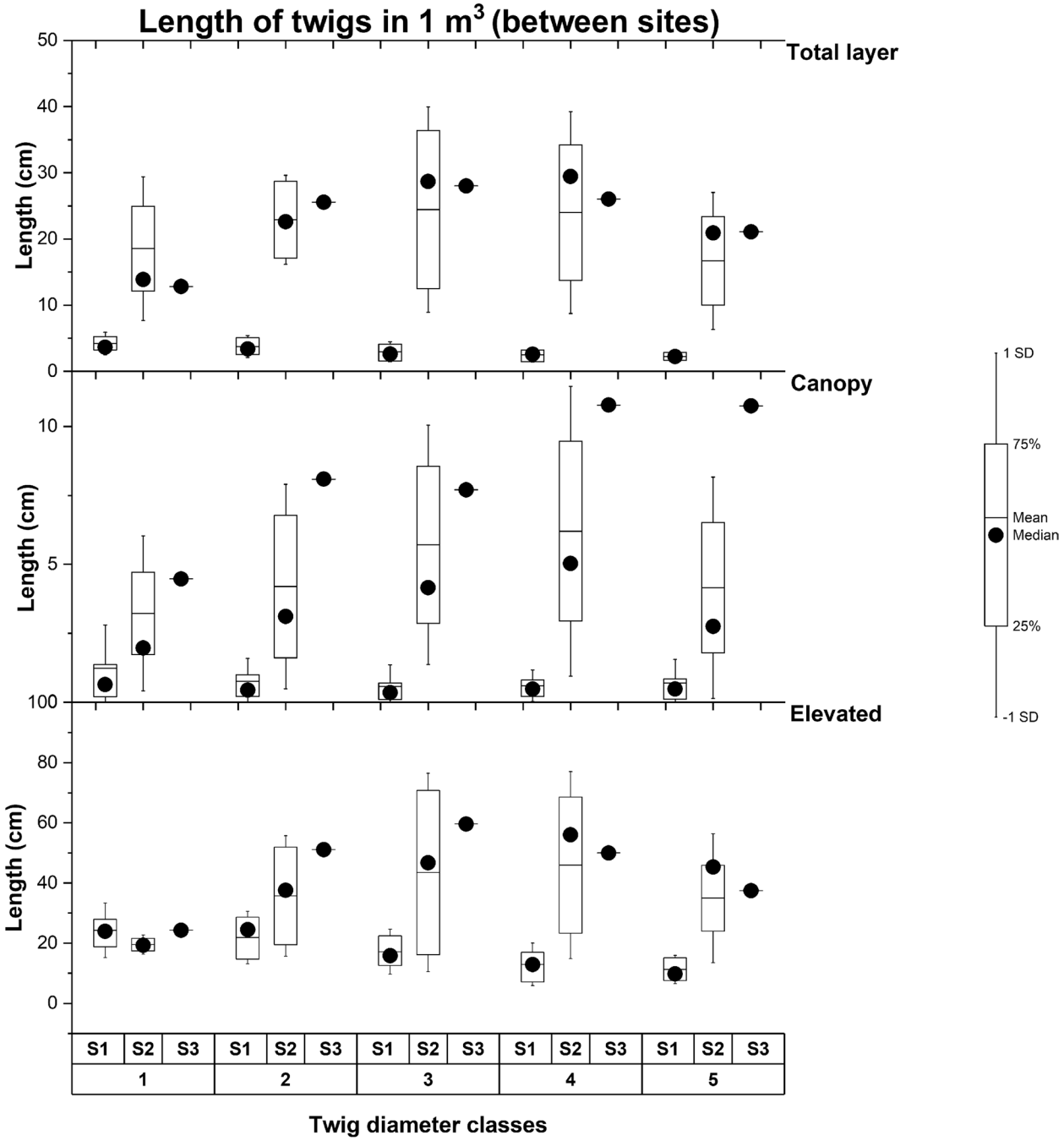


Fig. 9 Length of twigs for different diameter classes in 1 m^3. S1, S2, and S3 are Otway, Wombat, and Stony sites, respectively. Twig diameter classes are as follows: 1 (from 0.51 to 1.5 cm), 2 (from 1.51 to 2.5 cm), 3 (from 2.51 to 3.5 cm), 4 (from 3.51 to 4.5 cm), and 5 (from 4.51 to 5.5 cm). Otway (S1) has 24 plots, Wombat (S2) has 4 plots, and Stony (S3) has 1 plot

decrease from class 3 to class 5, indicating a general trend of decreasing twig length with increasing diameter. The Elevated layer has the longest mean twig lengths, ranging from 15.4 to 24.8 cm.

The Canopy layer shows relatively homogeneous mean twig lengths across all diameter classes, with values ranging from 1.5 to 1.7 cm and a similar trend for standard deviation, with values from 0.5 to 0.8 cm.

The Total layer presents a similar pattern to the Elevated layer, with an initial increase in mean twig length from class 1 to class 2, then a decrease from class 2 to class 5. However, the Total layer has shorter mean twig lengths compared to the Elevated layer. The Canopy layer

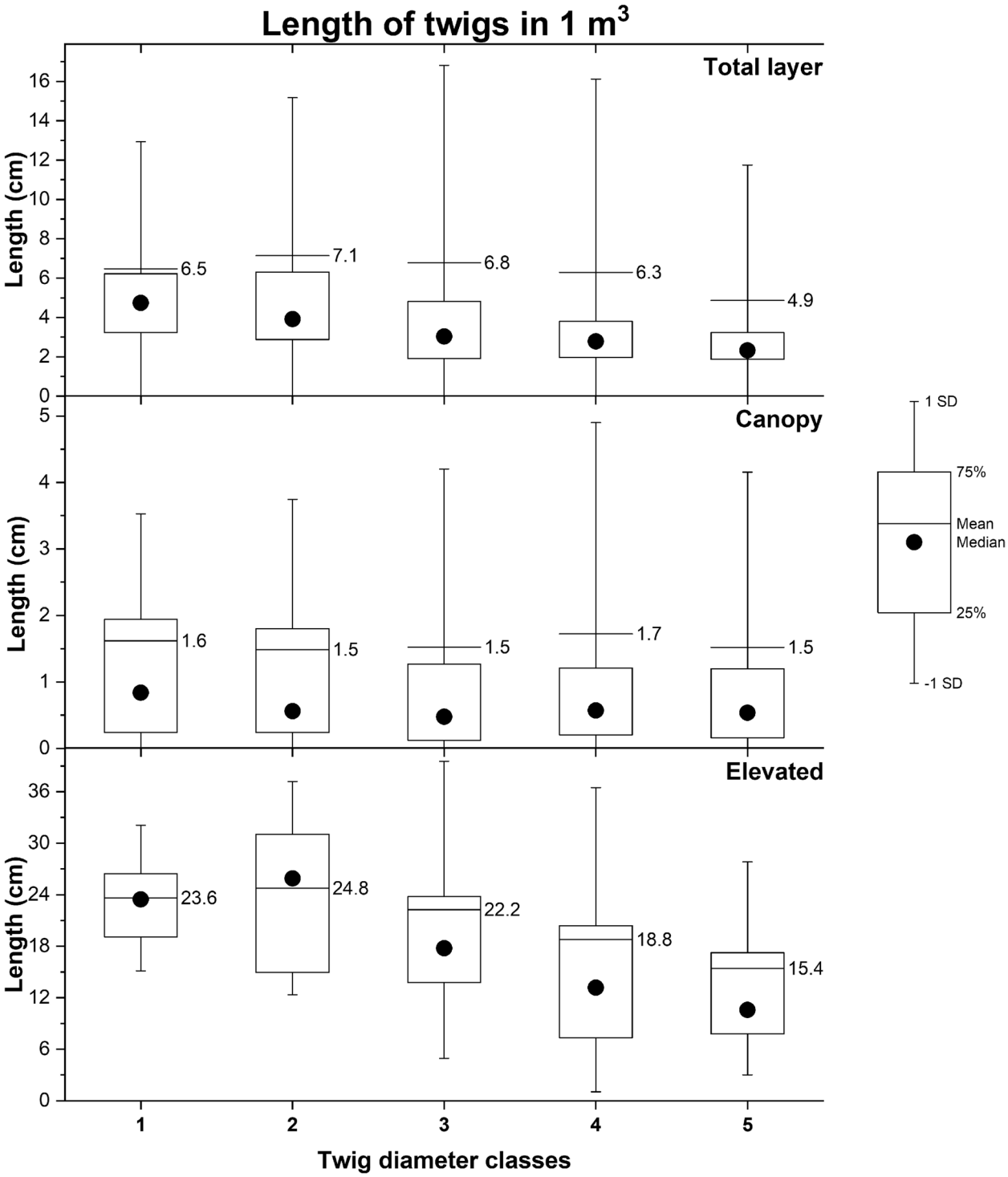


Fig. 10 Length of twig diameter classes in 1 m^3 for each layer. Twig diameter classes are as follows: 1 (from 0.51 to 1.5 cm), 2 (from 1.51 to 2.5 cm), 3 (from 2.51 to 3.5 cm), 4 (from 3.51 to 4.5 cm), 5 (from 4.51 to 5.5 cm). The boxplot symbol on the right explains how to read the main figure

consistently has the smallest mean twig lengths across all diameter classes.

All three layers show similar patterns in their maximum values, all peaking between 2.51 and 4.5 cm twig diameter (class 3 and class 4). Maximum twig lengths in the Total layer increase from 13 to 16.6 cm (class 3) and then decrease to 11.5 cm. In the Canopy layer, maximum twig lengths increase from 3.6 to 4.9 cm (class 4) and then decrease to 4.1 cm. In the Elevated layer, maximum twig lengths increase from 32 to 39.5 cm (class 3), then decrease to 28 cm.

Candlebark

Figures 12 and 13 show variations in candlebark length across different width classes and layers (Total, Canopy, Elevated) in 1 m^3. The candlebark width is categorized into three classes ranging from 0.51 to 15 cm. The candlebark metrics include mean, median, quartiles and range of width classes and length.

Figure 11 compares candlebark lengths between the three classes. In the Total layer, the mean length is higher for the smallest width class (class 1) and decreases with increasing width. Consequently, the smallest mean length is found in the largest width class (class 3). This distribution pattern is consistent across the Canopy and Elevated layers, where candlebark length decreases with increasing width. Similar characteristics are observed for the maximum candlebark lengths.

Although the pattern of candlebark length distribution over the width classes is similar for all layers, there are significant differences in the range of candlebark lengths among the layers. The largest length range is observed in the smallest width class of the Elevated layer (10–40 cm), followed by the Total (7–20 cm) and the Canopy layers (0–6 cm).

Generally, when comparing sites in different layers and candlebark widths, means and medians of the length of candlebark in the Otway site (S1) remain the lowest

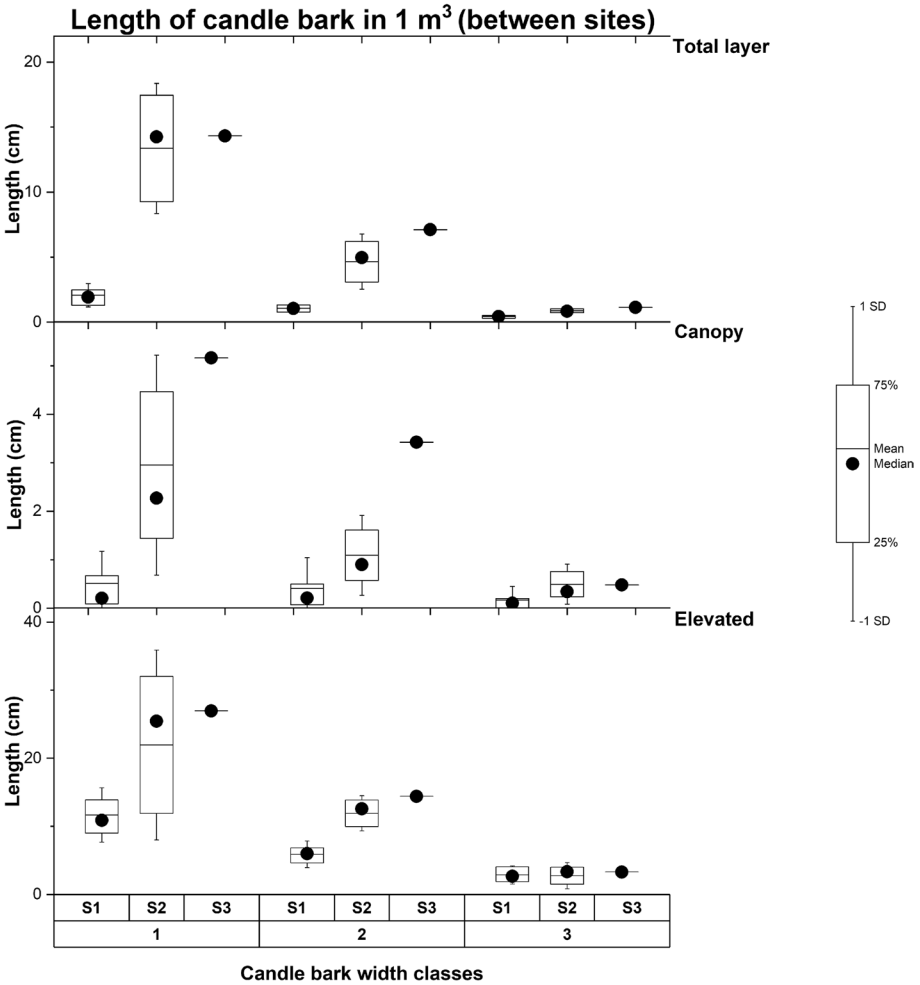


Fig. 11 Length of candlebark versus width classes in 1 m^3. S1, S2, and S3 are the Otway, Wombat, and Stony sites, respectively. Candlebark width classes are as follows: 1 (from 0.51 to 5 cm), 2 (from 5.01 to 10 cm), and 3 (from 10.01 to 15 cm)

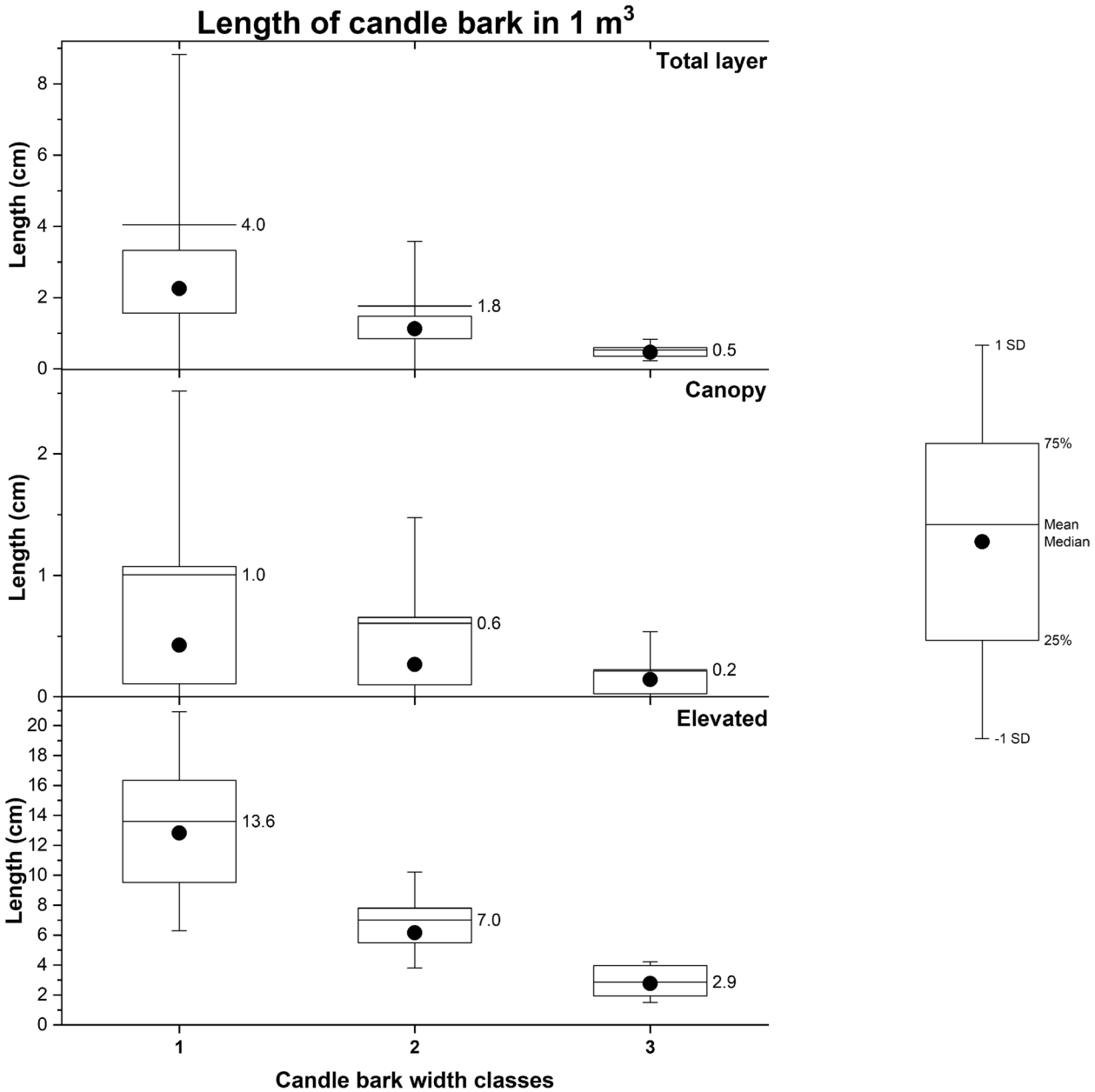


Fig. 12 Length of candlebark versus width classes in 1 m^3. Candlebark width classes are as follows: 1 (from 0.51 to 5 cm), 2 (from 5.01 to 10 cm), and 3 (from 10.01 to 15 cm)

across different layers (< 4 cm for the Total layer, < 1.2 cm for the Canopy layer, and < 16 cm for the Elevated layer). Means and medians of candlebark lengths in the Stony site (S3) are the highest (from 1 to 14 cm for the Total layer, from 0.5 to 5.2 cm for the Canopy layer, and from 3 to 28 cm for the Elevated layer).

Figure 12 shows lengths of candlebark width-class data for all three sites. We observed that mean bark lengths decrease as the bark width increases. This trend is consistent across all layers.

Overall, the Elevated layer demonstrates the greatest variability and largest candlebark lengths when

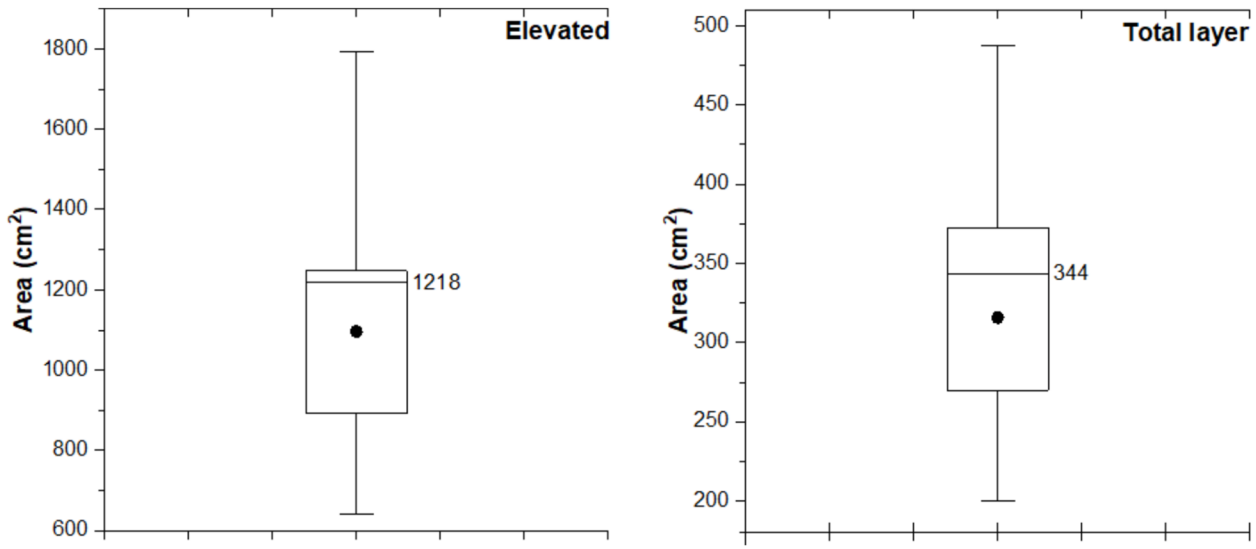


Fig. 13 Area of stringybark in 1 m^3 per fuel layer (Total layer—left, Elevated layer—right). The black dot in boxplot indicates the median value, line in boxplot indicates the mean value

comparing the different fuel layers, ranging from 1.8 to 22 cm. In contrast, the Canopy layer shows the shortest bark lengths, with much less variation, ranging from 0.1 to 2.5 cm. The Total layer shows a moderate level of variability and falls between the Elevated and Canopy layers in terms of both mean lengths and ranges, ranging from 0.3 to 8.9 cm. This pattern is consistent across width classes and layers and shows that most candlebark fuel falls within a length range of 0.5–10 cm in 1 m^3, with the Elevated layer contributing the most variability. The results also highlight the dominance of narrower candlebark and its significant variability.

Stringybark

Figure 13 shows the stringybark area (trunk area) in 1 m^3 for all 3 sites. Stringybark area also varied significantly across sites, indicating variability in tree size and potentially in the structure of the forest stand across different plots (Appendix B). The mean area of stringybark in the Elevated layer (1218 cm^2) is roughly 3.5 times higher than in the Total layer (344 cm^2).

When comparing data ranges, the Elevated layer demonstrates a much larger distribution, with stringybark areas ranging from approximately 630 to 1800 cm^2, while the Total layer shows a narrower range between 200 and 480 cm^2.

Discussion

Firebrand fuel characterisation method

Fire behavior prediction models rely on accurate fuel information as an input parameter (Gould et al. 2014; Tolhurst et al. 2008). Variation in forest types, local environments, and previous fire disturbances affects fuel amounts; the complexity of which poses a significant challenge for accurate fire behavior modeling. Variations in vegetation across regions and the development of different fire models have led to the use of varying methods for describing, measuring, and mapping fuels (Duff et al. 2017). Usually, remote sensing methods (D'Este et al. 2021; Kearney et al. 2022; Sesnie et al. 2018; Wells et al. 2023) or ground measurements (Gould et al. 2011) are used to describe the fine fuel as mass per hectare (kg/ha or t/ha). These methods often lack accuracy or do not provide information about firebrand fuel.

The method presented in this study offers a preliminary approach for reconstructing 3D forest structure with high accuracy and resolution, enabling fuel quantification at a fine spatial scale within a 1 m^3 volume. The potential advantage of this method lies in its ability to determine the length of twigs and bark (cm/m^3) for a particular diameter class across different forest layers. This may provide insights into fine fuel loads for each vegetation layer and could serve as a preliminary framework for future research on firebrand generation at a laboratory scale. While this study is based on a limited number of

sites, the goal was to develop a new method for categorizing variations in firebrand fuel, which may be useful for future firebrand generation experiments. Further validation with larger datasets is needed before broader application.

Forest type

The findings suggest differences in firebrand fuel characteristics across the three sites. The Wombat and Stony sites had much higher maximum twig lengths (13–36 cm in Total layer) and maximum candlebark lengths (2–18 cm in Total layer) compared to the Otway site (its maximum twig lengths are < 6 cm, and candlebark lengths are < 3 cm) (Figs. 9 and 11), despite the similar types of *Eucalyptus* forests. Consequently, the Wombat and Stony sites may have a greater potential to generate firebrands than the Otway site under similar fire conditions. This suggests that variations in environmental conditions may play an important role in producing firebrand fuel. This is a critical factor in understanding fire spread dynamics and developing effective fire management strategies in these areas.

The twig length and diameter of Wombat and Stony sites appear similar (Fig. 9), which may be a consequence of similar rainfall and topography. The trend in twig length at Wombat and Stony increases from twig diameter class 1 to class 4, and decreases in class 5. In contrast, Otway shows a decreasing trend as the diameter increases (Fig. 9). The imbalance in the number of plots per site may account for observed differences between sites. The overall trend across the three sites shows a decrease in twig length with increasing diameter (Fig. 10). However, other studies (Basile and Hutchings 1966; Xiang et al. 2009) reported an opposite trend between twig diameter and length. The variability observed here likely reflects the heterogeneity of forest structure and composition, and further investigation is needed.

Fuel layers

The Elevated layer appeared to contain the largest amount of firebrand fuel, including longer twigs, candlebark, and a larger area of stringybark, followed by the Total and Canopy layers. This may be partly attributed to the smaller volume of the Elevated layer compared to the other layers, as all metrics were normalized by layer volume to obtain values in $1m^3$. Furthermore, the Elevated layer at our sites comprises shrubs and small trees; these typically have a higher density of small twigs, similarly described in other forests (e.g., Götmark et al. 2016; Mori et al. 2024).

The variability in stringybark area in the Elevated layer (Fig. 13) highlights differences in bark thickness, which generally increases with tree girth or stem diameter. This trend is consistent across various species and environments (e.g., Hedge et al. 1998; Paine et al. 2010; Rosell 2016; Shearman and Varner 2021). The maximum stringybark area observed in the Elevated layer (1800 cm^2) far exceeds that of the Total layer (480 cm^2), underscoring the substantial differences in bark characteristics between layers. Most firebrand characterization studies have focused on burning entire tree samples, without consideration of different tree parts, leading to inconsistent sampling methods and a lack of quantitative guidance on sample size (Adusumilli and Blunck 2023; Almeida et al. 2021). However, in wildland fires, firebrands are often reported to predominantly consist of bark scales originating from trunks, which may account for over 70% of firebrand material (El Houssami et al. 2016). By estimating the area of stringybark along a trunk, this study provides preliminary insights into fuel quantity that could inform firebrand experiments.

Firebrand fuel

Our results indicate that generally, as twig diameter increases, twig length tends to decrease. This is consistent with the findings of Raumonen et al. (2013). However, we found that the mean length of 1 cm diameter twigs was slightly lower than that of 2 cm diameter twigs (Fig. 10), which was unexpected. This result may be explained by the fact that smaller twigs are often obscured by larger twigs, thereby preventing them from being detected by the laser pulse during TLS scans. While TreeQSM has been found to be reliable with a low measurement bias (Demol et al. 2022a, b), there are challenges to accurately capturing small fuel elements using TLS.

For candlebark, a similar inverse relationship is observed, with mean bark length decreasing as the bark width increases. This trend is consistent across all layers and may be attributed to the natural process of bark shedding in strips, which are typically narrow. Longer candlebark lengths in the Elevated layer compared to the Canopy layer are likely due to bark peeling from trunks and wind carrying some bark to the ground. Environmental factors influencing bark size variation have been noted in previous studies (Pokswinski et al. 2020; Rosell et al. 2014).

Implications for fire behaviour modelling and fuel management

Although preliminary, these results highlight the potential for using TLS-derived metrics to inform firebrand

generation studies and fire behavior models. Linking structural attributes of forests to fire behavior could eventually support targeted management strategies, such as selective thinning or prescribed burns in high-risk areas. The proposed 1 m^3 volumetric framework may offer a practical basis for improving fuel characterization, but further validation and refinement are essential. Future work should also explore integration of ground-based TLS with aerial LiDAR to develop high-resolution fuel databases at broader scales.

Limitations

The three study sites had different numbers of plots—Otway (24 plots), Wombat (4 plots), and Stony (1 plot). This limited the ability to make robust site-level comparisons. Future studies could address this by incorporating a larger and more balanced number of plots across sites to enable robust statistical comparisons.

The calibration results for candlebark showed that the best relative accuracy between field measurements and TreeQSM measurements at 20° was about 10%. However, our result is based on a single tree, and a larger set of samples may yield different values. Also, TreeQSM does not identify individual candlebark elements, so we compared the total length of all candlebark elements by applying the algorithm to one tree. Future studies should focus on expanding this method to a larger number of samples and developing approaches to separate individual bark elements for more detailed analysis.

When processing the TLS data, we experienced difficulties in accurately extracting individual trees from the point cloud using the Treeiso algorithm, especially when the forest structure was dense rather than sparse. This led to the requirement for manual segmentation to ensure accurate TreeQSM reconstruction. Superior algorithms or additional techniques will be required to improve accuracy in dense forest conditions.

Differences in scanning equipment may also have affected the accuracy of our results. The Otway site was scanned with the FARO Focus 3DS, while the Wombat and Stony plots were scanned with the more advanced RIEGL VZ-400i. We observed that the density of points in the datasets from Otway was lower than in the other sites. Future studies should utilize maximum point cloud density and assess the influence of scanning equipment on accuracy.

In this study, we did not distinguish between candlebark and stringybark within each plot, as our primary aim was to determine the maximum firebrand fuel ranges for future firebrand generation experiments. However, this may be estimated if the proportion of stringybark and candlebark trees at a given plot is known.

Conclusion

This study presents a framework for characterizing fine-fuel distributions across vegetation communities, with a focus on firebrand generation in wildfires. Using terrestrial laser scanning and TreeQSM, fine-fuel distributions were quantified across forest layers within a standardized 1 m^3 volume. Twig and bark dimensions varied significantly with height, with the greatest firebrand fuel presence occurring in the near-surface and elevated layers and the lowest in the canopy. Twig length decreased with increasing diameter, and candlebark length decreased with increasing width, highlighting the importance of vertical fuel structure in controlling firebrand availability.

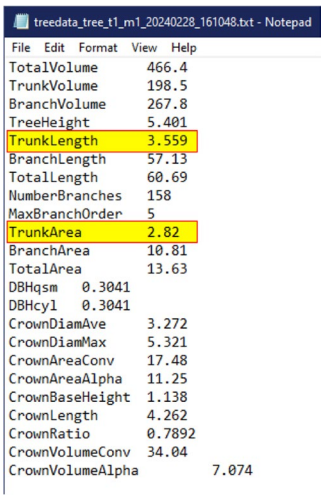
TLS proved to be a promising tool for high-resolution estimation of vegetative fuel quantity and strata volume, particularly for fine fuels within forest layers. The observed variation in potential firebrand fuels among layers and between sites suggests that some forests may have a greater capacity to produce firebrands, potentially contributing to spotting-driven fire spread. However, these interpretations should be treated with caution given the limited sample size, underscoring the need for broader validation across additional vegetation types.

Medium- and long-range firebrand transport remains poorly captured in both surface and coupled fire-atmosphere fire spread models. By quantifying potential firebrand source fuels across forest layers, the present framework establishes a foundation for improved representation of firebrand generation processes. The inclusion of firebrand information that has been generated from experiments can improve the realism of fire spread models, which will result in improved predictions of the spotting contribution to wildfire spread. Continued validation and refinement will be essential to ensure the robustness of TLS-based firebrand fuel characterization and its suitability for integration into spotting modules within fire behavior models.

Appendix A
Site and plot information

Site and scanner	Plot	Max tree height (m)	Canopy base height (m)	Plot size (L × W) (m)	
Otway FARO Focus 3DS	S057	26	9.1	20	20
	S060	26.1	14	20	20
	S084	35	16	20	20
	S085	19.5	9.3	20	20
	S094	27.7	11.2	20	20
	S095	25.5	14.7	20	20
	S101	41.7	9	20	20
	S104	40	12	20	20
	S118	41	11.6	20	20
	S119	36	11.6	20	20
	S126	37.3	22	20	20
	S184	38.3	17.8	20	20
	SP01	25.8	11.5	20	20
	SP02	31.8	9	20	20
	SP03	31.3	7.6	20	20
	SP04	32.3	25.8	20	20
	SP06	25.8	8.3	20	20
	SP08	33.2	15.6	20	20
	SP09	17.1	7.9	20	20
	SP10	24	10	20	20
	SP11	23.3	10	20	20
	SP12	32.2	9	20	20
	SP13	39	11.2	20	20
	SP14	34.5	14.5	20	20
Wombat Riegl VZ-400i	Wombat site 1	29.7	18.2	34.6	29.4
	Wombat site 2	28.5	14.8	17	29
	Wombat site 3	29	15.4	26.4	24.8
	Wombat site 4	30	10.3	37	25.4
Stony Riegl VZ-400i	SEV03	15.4	5.5	7.4	8.2

Appendix B
An example of a metric output file derived from TreeQSM



a) Tree data information, *TrunkLength (m) is the height of trunk, TrunkArea (m^2)* is the area of trunk. We used these parameters to characterize the stringybark-firebrand fuel.



Source.Name	order	parent	diameter (m	volume (L	area (m^2)	length (m	height (m	angle (deg	azimuth (deg	zenith (deg
branch_tree_t1_m1_20231212_083535.txt	0	0	0.1258	45.1022	1.6462	5.1	0	0	15.4	32.7
branch_tree_t1_m1_20231212_083535.txt	1	1	0.0775	51.3001	2.4782	9.711	1.742	97.7	-63.5	75.3
branch_tree_t1_m1_20231212_083535.txt	1	1	0.0635	11.1273	0.6508	3.166	2.135	14.1	-28.6	63.9
branch_tree_t1_m1_20231212_083535.txt	2	5	0.0614	1.8742	0.1405	0.905	2.19	27.4	-78.7	126.2
branch_tree_t1_m1_20231212_083535.txt	2	5	0.0443	2.6054	0.1909	1.126	2.254	38.8	-41.8	60.7
branch_tree_t1_m1_20231212_083535.txt	2	5	0.0967	31.0421	1.2838	4.225	2.544	62.5	138.8	56.7
branch_tree_t1_m1_20231212_083535.txt	2	5	0.0812	2.8973	0.1456	0.602	3.009	165.4	86.1	121.6
branch_tree_t1_m1_20231212_083535.txt	2	5	0.0614	1.0162	0.0832	0.593	2.735	115.4	-72.3	98.4
branch_tree_t1_m1_20231212_083535.txt	2	5	0.0931	12.7128	0.6609	3.167	2.799	20.1	5	108
branch_tree_t1_m1_20231212_083535.txt	2	5	0.0656	1.2943	0.0861	0.46	2.82	132.4	76.4	84.5
branch_tree_t1_m1_20231212_083535.txt	2	5	0.0514	0.6568	0.0541	0.364	2.953	122.6	37.3	69.4
branch_tree_t1_m1_20231212_083535.txt	2	5	0.0166	0.0971	0.0234	0.447	2.964	142.1	-110.2	101.2
branch_tree_t1_m1_20231212_083535.txt	2	5	0.053	1.2758	0.1146	0.847	2.953	69.8	-120.1	136.6
branch_tree_t1_m1_20231212_083535.txt	2	5	0.2242	139.292	2.4848	3.527	3.076	139.5	98	76
branch_tree_t1_m1_20231212_083535.txt	2	5	0.0838	3.5529	0.1614	0.584	3.178	103.1	145.8	83.2

b) Branch-level tree information generated by TreeQSM. Where *Source.Name* refers to the text file associated with each branch; *order* indicates the hierarchical level of the branch relative to the main trunk; *parent* identifies the segment from which the branch originates; *diameter (m)* represents the branch diameter; *volume (L)* is the branch volume; *area (m^2)* is the branch surface area; *length (m)* is the branch length; and *height (m)* is the elevation of the branch from the ground. *Angle (deg)* denotes the angle between the branch and its parent; *azimuth (deg)* is the horizontal orientation; and *zenith (deg)* is the vertical inclination.

Appendix C

Twigs, candlebark, and stringybark distribution in 1 m^3

Plot	Layer	Twigs (cm)					Candlebark (cm)			Stringybark (cm^2)
		1	2	3	4	5	1	2	3	
Otway S057	Total layer	5.49	6.84	5.25	3.93	3.32	2.82	1.02	0.47	326.62
	Canopy	0.84	1.27	0.42	0.54	0.70	0.63	0.24	0.21	
	Elevated	32.03	31.16	23.77	15.51	17.24	14.25	5.53	2.77	887.77
	Midstory	7.33	11.89	10.59	8.22	4.74	4.09	1.32	0.25	
Otway S060	Total layer	5.09	5.79	5.11	4.95	3.62	3.32	1.25	0.47	277.805
	Canopy	0.83	0.16	0.07	0.28	0.48	0.06	0.32	0.10	
	Elevated	29.18	32.18	30.88	32.06	22.89	19.86	7.10	3.65	1098.58
	Midstory	4.34	5.98	4.82	3.97	2.73	3.17	0.95	0.16	
Otway S084	Total layer	4.90	2.88	1.92	2.00	2.03	1.94	1.15	0.54	836.93
	Canopy	1.84	0.90	0.75	0.68	0.97	0.91	0.66	0.17	
	Elevated	18.04	11.06	9.11	4.28	6.56	6.73	4.34	2.08	3164.87
	Midstory	6.78	4.15	2.23	3.42	2.67	2.51	1.27	0.76	
Otway S085	Total layer	4.96	2.88	1.90	2.12	2.93	1.72	2.31	1.45	292.54
	Canopy	5.94	2.19	1.27	2.04	2.88	1.85	2.94	1.36	
	Elevated	8.62	3.67	4.33	1.62	4.04	3.05	2.50	4.29	402.38
	Midstory	2.15	3.62	1.95	2.43	2.61	1.03	1.28	0.52	
Otway S094	Total layer	3.26	3.92	2.81	2.95	2.19	1.91	1.13	0.35	319.73
	Canopy	0.62	1.00	0.49	0.57	0.71	0.19	0.14	0.00	
	Elevated	18.43	17.21	13.78	13.16	9.24	10.55	6.56	2.77	1214.24
	Midstory	3.91	5.63	4.07	4.53	2.97	2.68	1.43	0.32	
Otway S095	Total layer	5.47	3.93	3.92	3.07	2.79	2.40	1.48	0.61	445.89
	Canopy	4.42	0.99	1.00	0.58	1.20	0.86	0.63	0.32	
	Elevated	12.01	8.59	11.34	11.12	7.79	7.81	7.81	1.75	1202.71
	Midstory	5.06	5.57	4.99	3.63	3.17	2.66	0.93	0.62	
Otway S101	Total layer	2.97	2.12	1.54	0.94	0.87	1.07	0.62	0.28	177.62
	Canopy	0.06	0.20	0.07	0.05	0.11	0.04	0.02	0.00	
	Elevated	33.54	21.42	15.94	5.12	6.24	10.24	5.53	3.25	1038.2
	Midstory	5.83	4.36	3.44	3.83	2.67	2.71	1.73	0.57	
Otway S104	Total layer	2.86	2.70	1.37	0.97	0.83	1.11	0.78	0.19	315.78
	Canopy	0.18	0.05	0.01	0.01	0.09	0.02	0.01	0.00	
	Elevated	32.43	34.00	17.76	11.13	8.30	12.82	7.58	1.78	1828.06
	Midstory	2.99	2.26	1.07	1.13	1.06	1.26	1.26	0.32	
Otway S118	Total layer	1.94	1.47	1.26	1.07	1.17	0.88	0.52	0.22	194.02
	Canopy	0.14	0.08	0.04	0.11	0.04	0.01	0.05	0.20	
	Elevated	20.99	16.28	15.17	8.44	11.53	9.69	6.43	0.85	1025.17
	Midstory	2.52	1.90	1.39	2.15	1.98	1.28	0.43	0.12	
Otway S119	Total layer	3.23	3.35	2.28	1.96	1.89	1.56	0.79	0.49	269.78
	Canopy	0.58	0.22	0.12	0.20	0.10	0.16	0.11	0.10	
	Elevated	21.67	24.64	14.47	12.83	13.44	11.20	7.31	4.77	1419.94
	Midstory	5.30	5.90	4.72	3.69	3.54	2.68	0.80	0.38	
Otway S126	Total layer	1.66	1.61	1.25	1.29	1.11	0.98	0.49	0.24	201.32
	Canopy	0.19	0.43	0.64	0.87	0.22	0.11	0.02	0.00	
	Elevated	13.40	12.39	6.58	3.99	7.41	7.47	3.82	2.08	949.27
	Midstory	1.32	1.16	1.04	1.28	0.99	0.83	0.44	0.20	

Plot	Layer	Twigs (cm)					Candlebark (cm)			Stringybark (cm^2)
		1	2	3	4	5	1	2	3	
Otway S184	Total layer	3.23	3.21	2.56	2.35	1.96	1.90	0.92	0.39	314.13
	Canopy	0.24	0.24	0.11	0.20	0.20	0.12	0.19	0.00	
	Elevated	26.44	27.18	22.39	20.39	15.22	13.47	6.16	4.83	1246.74
	Midstory	3.44	3.28	2.60	2.28	2.16	2.39	1.04	0.20	
Otway SP01	Total layer	3.42	4.95	4.66	3.60	2.79	2.26	1.33	0.41	363.48
	Canopy	0.89	0.79	0.47	1.21	0.47	0.50	0.32	0.20	
	Elevated	17.37	28.42	28.50	16.71	9.73	10.17	7.83	1.94	1246.74
	Midstory	3.58	5.05	4.69	3.75	4.56	2.86	1.14	0.32	
Otway SP02	Total layer	3.83	3.40	2.35	2.83	2.38	2.30	1.34	0.54	289.78
	Canopy	0.34	0.26	0.14	0.25	0.16	0.17	0.08	0.02	
	Elevated	28.29	27.19	18.57	21.42	17.74	18.38	11.66	3.82	1422.84
	Midstory	6.70	5.26	3.88	4.75	4.28	3.62	1.77	1.09	
Otway SP03	Total layer	4.74	3.96	2.69	2.93	2.24	1.98	0.97	0.49	259.53
	Canopy	1.71	1.80	1.08	1.41	1.16	0.71	0.45	0.20	
	Elevated	23.44	14.93	13.76	14.71	9.25	9.97	4.88	3.97	784.46
	Midstory	9.66	8.60	4.74	4.24	3.86	3.94	1.44	0.14	
Otway SP04	Total layer	3.83	3.81	3.03	2.78	2.32	2.26	1.09	0.47	372.49
	Canopy	0.16	0.25	0.15	0.20	0.11	0.05	0.05	0.00	
	Elevated	26.44	27.18	22.39	20.39	15.22	13.47	6.16	4.83	1246.74
	Midstory	2.42	2.29	1.76	1.62	1.56	1.67	0.84	0.13	
Otway SP06	Total layer	5.41	5.16	4.23	3.39	2.73	2.56	0.64	0.32	348.74
	Canopy	0.96	0.56	0.53	0.53	0.54	0.46	0.10	0.04	
	Elevated	27.49	28.74	23.15	14.90	14.96	13.51	4.27	1.77	1094.05
	Midstory	9.32	8.87	7.22	7.05	4.06	4.16	0.70	0.54	
Otway SP08	Total layer	3.24	3.21	3.16	2.75	1.87	1.89	1.09	0.51	220.46
	Canopy	0.56	0.44	0.42	0.74	0.35	0.43	0.27	0.21	
	Elevated	22.88	31.04	31.97	21.26	9.77	15.88	5.87	4.64	874.2
	Midstory	3.08	1.63	1.35	1.93	2.40	1.18	1.27	0.12	
Otway SP09	Total layer	7.49	7.34	6.40	4.34	3.15	4.04	1.66	0.59	301.75
	Canopy	3.49	2.02	1.61	1.03	2.13	1.40	0.91	0.47	
	Elevated	24.62	25.88	21.47	12.96	6.59	13.03	6.07	2.12	667.89
	Midstory	6.39	7.81	7.58	5.99	3.31	4.39	0.91	0.10	
Otway SP10	Total layer	8.52	5.20	4.00	3.80	3.64	3.22	1.45	0.38	345.52
	Canopy	0.65	0.30	0.27	0.25	0.58	0.22	0.54	0.03	
	Elevated	54.77	24.40	15.71	17.13	18.56	16.34	5.90	2.51	2704.06
	Midstory	7.80	7.97	7.06	5.98	4.37	4.45	1.67	0.33	
Otway SP11	Total layer	6.22	6.31	4.81	2.38	3.23	3.95	1.30	0.21	207.47
	Canopy	3.72	3.34	3.64	1.99	3.05	2.75	1.39	0.14	
	Elevated	26.19	32.02	19.87	6.82	8.83	16.35	2.86	0.73	468.81
	Midstory	4.00	3.00	1.85	1.59	1.69	1.93	0.64	0.16	
Otway SP12	Total layer	3.47	1.84	1.34	1.20	1.88	1.27	0.90	0.52	273.05
	Canopy	1.02	0.52	0.22	0.42	0.48	0.36	0.23	0.07	
	Elevated	20.51	10.24	6.66	6.38	10.56	8.45	5.49	4.15	1711.47
	Midstory	5.64	3.35	3.28	1.96	3.51	1.73	1.53	0.74	
Otway SP13	Total layer	3.44	2.36	1.28	1.59	1.46	1.34	0.85	0.36	237.03
	Canopy	0.20	0.10	0.02	0.17	0.07	0.11	0.12	0.13	
	Elevated	24.41	19.87	8.84	7.34	11.94	9.51	6.30	2.55	1533.31
	Midstory	7.75	4.57	3.16	4.45	2.89	2.92	1.62	0.47	

Plot	Layer	Twigs (cm)					Candlebark (cm)			Stringybark (cm^2)
		1	2	3	4	5	1	2	3	
Otway SP14	Total layer	2.06	1.78	1.59	1.22	0.98	0.96	0.42	0.17	276.41
	Canopy	0.03	0.17	0.10	0.06	0.01	0.07	0.02	0.03	
	Elevated	19.15	14.48	15.41	10.97	6.80	7.69	3.12	0.71	1237.62
	Midstory	1.88	1.82	1.20	1.12	1.37	1.04	0.53	0.31	
Wombat plot 1	Total layer	15.25	27.98	36.18	33.37	23.38	17.69	6.83	1.13	477.35
	Canopy	1.52	2.09	3.19	4.73	2.52	1.81	0.83	0.27	
	Elevated	23.62	55.88	75.17	68.89	46.15	34.19	14.10	3.64	1237.22
	Midstory	23.98	42.51	54.13	48.69	35.04	26.70	10.07	1.35	
Wombat plot 2	Total layer	11.75	17.25	21.15	25.56	18.42	11.29	4.34	0.95	496.18
	Canopy	7.42	9.43	12.01	13.59	10.05	6.20	2.25	0.41	
	Elevated	19.47	27.38	27.03	43.82	45.49	21.10	11.54	4.33	1096.04
	Midstory	14.99	23.90	30.14	35.17	22.25	14.95	5.21	0.86	
Wombat plot 3	Total layer	12.51	29.42	36.62	35.05	23.37	17.21	5.60	0.73	319.14
	Canopy	1.94	4.13	5.11	5.34	2.97	2.73	0.97	0.20	
	Elevated	19.08	47.85	66.37	68.27	45.21	29.81	13.60	3.01	891.43
	Midstory	22.37	52.52	64.08	59.93	40.64	30.05	8.92	0.85	
Wombat plot 4	Total layer	12.81	25.55	28.02	26.03	21.08	14.32	7.11	1.14	405.69
	Canopy	4.47	8.09	7.70	10.78	10.75	5.16	3.42	0.48	
	Elevated	24.30	51.10	59.61	49.97	37.44	26.96	14.41	3.29	994.47
	Midstory	30.18	61.44	69.22	56.89	41.92	33.40	14.10	2.10	
Stony SEV03	Total layer	34.67	16.92	3.80	1.98	1.58	7.27	1.84	0.71	700.09
	Canopy	2.00	1.13	2.53	1.16	1.06	1.07	0.31	1.10	
	Elevated	15.83	11.61	5.27	2.65	2.79	2.70	8.38	0.00	854.62
	Midstory	158.19	73.44	6.77	4.10	2.30	31.52	1.43	0.00	

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Authors' contributions
HN, AF, TP, MP, JF, and AAP conceived the ideas and hypotheses and designed the methodology. SP, NH, and HN led the LiDAR data collection. HN and AF conducted the TLS analysis of the data. All authors contributed to interpreting results. HN drafted the manuscript, and all authors contributed editorial input during manuscript preparation. All authors read and approved the final manuscript.

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Data availability
The TLS datasets used in this study were provided by third parties and are subject to their data-sharing agreements; therefore, they are not publicly available and require approval from the data owners for access. Data products and findings derived from the processing and analysis of these datasets are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

Author details
[1]Faculty of Science, The University of Melbourne, 4 Water Street, Creswick, VIC 3363, Australia. [2]Centre National de la Recherche Scientifique (CNRS), Sciences Pour l'Environnement – Unité Mixte de Recherche 6134, Università di Corsica, Campus Grossetti, Corte 20250, France. [3]Australian Government Bureau of Meteorology, Melbourne, VIC, Australia.

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